

Vinh Ly

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Summary

Software Engineer with **18 years** of end-to-end ownership of high-traffic services that serve millions of daily users. Designed and operated **C++**, **Java**, **Go**, **Python** and **Kubernetes** based platforms supporting real-time bidding, video streaming, and marketplace transactions. Delivered a latency-reduction project that cut request time by **27 %** while handling a 3× traffic surge. Partnered closely with product, security, and data-science teams to align feature delivery with business goals. Next focus is on cloud-native deployment automation and on-call reliability for multi-region services. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages & Scripting: TypeScript, JavaScript, HTML, CSS, SQL

Work Experience

Education

B.S. Honors

Virginia Tech | 2004 - 2007

Staff Software Engineer, YouTube (Remote) 02/2025 - Present

- Architected a multi-region video ingestion pipeline using **Go** and **Kubernetes**, increasing throughput by **45 %** while maintaining sub-second processing latency.
- Implemented a **C++** transcoding service that reduced CPU usage by **22 %** through SIMD optimizations and kernel-level thread pooling.
- Led on-call rotation for the live-streaming stack, decreasing incident MTTR by **30 %** via automated root-cause analysis tools.
- Collaborated with product and data teams to launch a real-time recommendation engine, delivering **1.2 B** additional impressions in the first quarter.
- Mentored a cross-functional squad on performance testing best practices, raising code coverage of latency tests from 60 % to 95 %.

Staff Software Engineer, Whatnot (Remote) 05/2024 - 02/2025

- Built a **Python**-driven auction matching service that handled a peak of **12 M** concurrent bids, improving match accuracy by **18 %**.
- Refactored legacy **Java** order-processing modules into containerized microservices, cutting deployment time from hours to minutes.
- Introduced automated canary releases with **Terraform**, reducing rollback incidents by **40 %**.
- Partnered with security engineers to embed runtime policy enforcement, achieving compliance with PCI-DSS within three months.
- Optimized database query paths, decreasing average query latency from 120 ms to 68 ms.

Staff Software Engineer/Manager, Google (Remote) 03/2015 - 04/2024

- Directed a team of 12 engineers to deliver a global search indexing pipeline written in **C++**, scaling to **200 PB** of data.
- Introduced kernel-level I/O batching that lowered disk write latency by **15 %** across all data centers.

- Managed on-call responsibilities for critical authentication services, achieving 99.99 % availability over 3 years.
- Co-authored a **Go** based load-balancer that improved request distribution uniformity by **22 %**.
- Implemented a cross-team observability framework, reducing mean time to detection by **35 %**.

Software Engineer, Google (Remote) 02/2011 - 02/2015

- Developed a **Java** ad-serving platform that processed over **3 B** ad impressions daily, increasing fill rate by **9 %**.
- Optimized cache eviction policies, cutting cache miss rate by **12 %** and saving \$1.2 M in infrastructure costs.
- Created a **Python** monitoring agent that automated anomaly detection, reducing false alerts by **40 %**.
- Collaborated with product managers to launch a new bidding algorithm, boosting revenue per user by **5 %**.
- Performed kernel-level profiling to identify bottlenecks, leading to a 10 % CPU reduction.

Software Design Engineer II, Microsoft (Remote) 10/2007 - 12/2011

- Engineered a **C++** file-synchronization service for enterprise clients, supporting over **500 K** concurrent connections.
- Implemented end-to-end encryption using kernel-level cryptographic APIs, meeting FIPS 140-2 compliance.
- Automated build pipelines with **Jenkins**, decreasing integration time from 2 hours to 15 minutes.
- Worked with cross-functional teams to integrate the service into Azure Marketplace, generating \$3 M in first-year revenue.
- Reduced memory footprint of core libraries by **18 %** through careful allocation strategies.

Software Design Engineer Intern, Microsoft (Remote) 06/2006 - 08/2006

- Contributed to a **Python** prototype for automated log parsing, improving parsing speed by **30 %**.
- Wrote unit tests that increased code coverage from 45 % to 78 % for the assigned module.
- Collaborated with senior engineers to refactor legacy scripts, reducing runtime errors by **25 %**.